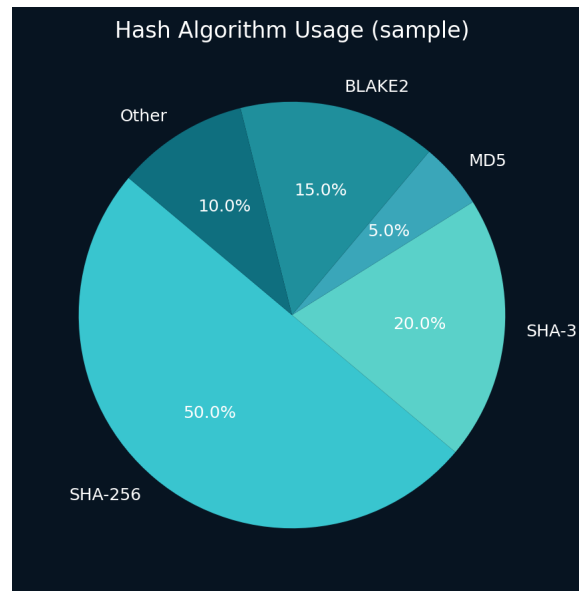


Hash Functions & Digital Signatures

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Attention: ONLY ACADEMIC PURPOSES. Use these materials in isolated labs and with permission.



Hash Functions

Hash functions map data to fixed-size digests. Use cryptographic hashes like SHA-256 or SHA-3 for integrity checks. Avoid MD5/SHA-1 for security-sensitive uses.

Digital Signatures

- RSA signatures, ECDSA, and EdDSA (Ed25519) for authentication and non-repudiation.
- Signatures require secure private key storage and proper canonicalization of data.

Use Cases

Code signing, document verification, and secure timestamping are common uses of hashes and signatures.